

MODULE 3.3

Account Management

Network Defense | Cisco Networking Academy

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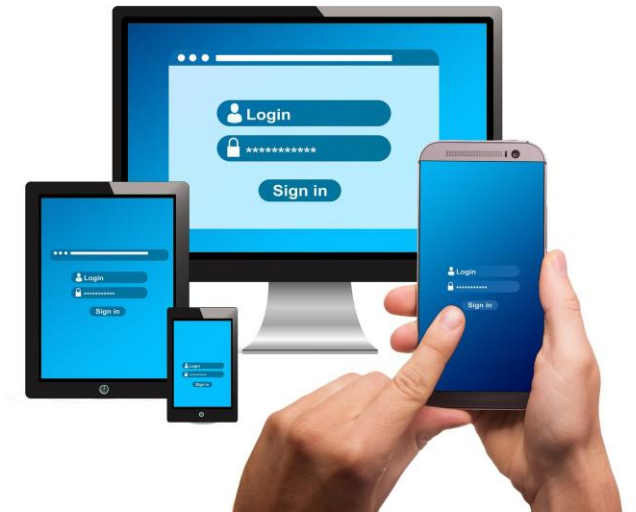
Topics:

- What is Account Management?
- Types of User Accounts
- File Access and Permissions
- Account Policies and Authentication
- Authentication Technologies and HMAC
- Summary

What Is Account Management?

- Creating, managing and protecting user accounts
- Controls who can access systems
- Protects sensitive information
- Tracks user activity

Example: Each student has their own login.



Types of User Accounts

Main Account Types

Standard User

- Uses everyday applications
- Cannot install software
- Cannot change important settings

Administrator

- Has full control
- Installs software
- Creates user accounts
- Changes system settings

Why is this important?

Using a standard account every day reduces security risks.



File Access and Permissions

File Permissions

Permissions decide what a user can do with files.

Common permissions:

- Read
- Write
- Modify
- Delete
- Full Control

Example

A teacher allows students to read assignment files but only the teacher can edit them.



Account Policies and Authentication

Account Policies

Rules that improve security.

Examples:

- Password length
- Password complexity
- Password expiration
- Account lockout after failed login attempts

Authentication

Authentication checks that a user is who they claim to be.

Examples:

- Password
- Fingerprint
- Face recognition
- One-Time Password (OTP)



Authentication Technologies and HMAC

Authentication Technologies

- Passwords
- Smart Cards
- Biometrics
- Multi-Factor Authentication (MFA)

HMAC

Hash-Based Message Authentication Code

HMAC checks that data has not been changed while being sent.

Simple Example

Online banking uses HMAC to make sure payment information is not altered.



Takeaways

- Different users have different account permissions.
- Authentication protects user identities.
- File permissions protect important data.
- Strong account policies improve security.
- HMAC helps verify that information remains unchanged.

References

Cisco Networking Academy

[Cisco Networking Academy: Learn Cybersecurity, Python & More](#)

Microsoft Learn

[Microsoft Learn: Build with answers in reach](#)

NIST Cybersecurity Framework

[Cybersecurity Framework | NIST](#)